

C. Michael
Haynes
03/30/26

Inverse Methods: HW 5 (EAS 6134 Problem Set 5)

Linear Inverse Problem: Data from Constable et al. (2018)

- 1) a. The overdetermined Least-Squares problem involves constructing a coarse "mesh" of points along the northing distance (i.e., along the track of the AUV). To do so, we consider steps of 40 meters, yielding 21 data points upon which we wish to determine the solution. For comparison, the data itself contains 33 measurements, with an average step of 24.5 meters.

We employ the normal equations solution as a first step here. This involves calculating $\vec{M}$ directly, via:

$$\vec{M}_{\text{OLS}} = (\underline{G}^T \cdot \underline{G})^{-1} \cdot \underline{G}^T \cdot \vec{d} \quad (1)$$

where $\vec{d}$ is the observed data (i.e., the $\vec{E}$ field) and $\underline{G}$ is a matrix that "averages" the two nearest data points around our (coarser) mesh. The solution to this inverse problem, via the normal equations (1), compared against the data as well as explicit extrapolation routines (nearest neighbor, linear interpolation between 2 nearest neighbors), are shown in the attached figure (following this page). The data, in black, include the noise associated with each measurement. The noise consists of stacking errors as well as a much larger (in magnitude) noise floor of 10 $\mu\text{V}/\text{m}$, employed by Constable et al. (2018) based on their knowledge of the instrumentation, measurement procedure, and any local anomalies. The stacking error is of order 0.1 $\mu\text{V}/\text{m}$, and is attributed with small variabilities in the actual measurements (i.e., not instrument-driven) that stem from normally-distributed, random, fluctuating signals.

The Normal equations solution, as seen in the figure (teal), captures the general behavior of the data but fails to fall within the noise floor compared to measurements in some locations. As indicated by the residuals, this solution is least effective near the strongest gradients in signal, and the residual plot demonstrates how the squared error grows quite large at these points.

1) b. To recast this problem as a linear inversion with more parameters than data, we construct a fine mesh, using a spacing of 10 meters, yielding 82 points upon which we wish to "interpolate" our data. We consider both a model norm minimizing solution, with the objective function

$$\vec{L}(\vec{m}) = \|\vec{m}\|^2 + \vec{\lambda}(\vec{d} - \underline{G} \cdot \vec{m}) \quad (2)$$

as well as a model-derivative norm minimizing solution,

$$\vec{L}(\vec{m}) = (\underline{L} \cdot \vec{m})(\vec{m} \cdot \underline{L}) + \vec{\lambda}(\vec{d} - \underline{G} \cdot \vec{m}) \quad (3) \text{ where}$$

$$\underline{L} = \begin{pmatrix} -1 & 1 & 0 & \dots & 0 \\ 0 & -1 & 1 & \dots & 0 \\ \vdots & 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \dots & -1 & 1 \end{pmatrix}$$

or represents a finite difference operator

to approximate the first derivative of the model. This will minimize the "roughness" of the model, rather than its magnitude. The analytical solutions to these inversions are included in a page following the plots. The two figures, representing each solution (orange: $\underline{L}_0$, or $\|\vec{m}\|$ minimizing; teal: $\|\underline{L}_1 \cdot \vec{m}\|$ minimizing) are included below.

As indicated by the figures, the solution for the Zeroth-order Tikhonov-regularized approach (2) is vastly over-fitted, i.e., since the model norm is a minimum, all values away from corresponding data points "spike" to nearly zero. However, as indicated by the residuals, the model does "fit" the data down to an accuracy of 10^{-29} MV/m, which certainly is below our computational limits even with double precision. However, the $\underline{L}_0$ model clearly does not represent any realistic physical scenario.

In contrast, the model-derivative norm minimizing solution, that is, Tikhonov First-order regularization, produces a far more realistic solution. That is because, as indicated by the signal, we have no reason to believe (physically-motivated) that the E field should be minimum magnitude outside of the observations. Rather, we would expect it to vary (on small scales) roughly quasi-linearly, which is enforced or targeted by including $\|\underline{L}_1 \cdot \vec{m}\|$ in our objective function. The residuals plotted below indicate the misfit is several orders of magnitude "worse" than the $\underline{L}_0$ solution, but is also several orders of magnitude below even the stacking noise levels, and well below the noise floor. This solution is the more physically-motivated one.

Both solutions were computed numerically with numpy's $\text{linv}()$ function. The model norm $\|\underline{L}_1 \cdot \vec{m}\| = 222.53$ MV/m, compared to $\|\vec{m}_{\text{OLS}}\| = 392.37$ MV/m. (For objective function (3))

Inverse Methods: Analytical Solutions

problem 1) b: underdetermined

$\Rightarrow$ need to include norm in objective function

$\rightarrow$ Start: Simply consider model norm:

$\vec{\lambda}$: vec of lagrange multipliers

$$L(\vec{M}, \vec{\lambda}) = \vec{M}^T \cdot \vec{M} + \vec{\lambda} \cdot (\vec{d} - \underline{G} \cdot \vec{M})$$

$$\partial_{\vec{M}} L = \underline{2\vec{M}} - \underline{G^T \cdot \vec{\lambda}} = 0 \quad (1)$$

$$\Rightarrow \vec{\lambda} = 2(\underline{G \cdot G^T})^{-1} \cdot \vec{d}, \text{ since}$$

(1) $\Rightarrow \vec{M} = \frac{1}{2} \underline{G^T \cdot \vec{\lambda}}$. substituting yields $\vec{\lambda}$,
which then produces

$$\vec{M} = \frac{1}{2} \underline{G^T \cdot \vec{\lambda}} = \frac{1}{2} \underline{G^T \cdot (2 \underline{G \cdot G^T})^{-1} \cdot \vec{d}}$$

$$\Rightarrow \vec{M} = \underline{G^T \cdot (\underline{G \cdot G^T})^{-1} \cdot \vec{d}}$$

$\rightarrow$ However, this is not physically motivated. Rather, we seek to minimize the "roughness", so we apply the 1st derivative model norm in our objective function:

$$\begin{aligned} L(\vec{M}, \vec{\lambda}) &= (\underline{L} \cdot \vec{M}) \cdot (\underline{L} \cdot \vec{M}) + \vec{\lambda} \cdot (\vec{d} - \underline{G} \cdot \vec{M}) \\ &= \vec{M} \cdot (\underline{L^T \cdot L}) \vec{M} + \vec{\lambda} \cdot (\vec{d} - \underline{G} \cdot \vec{M}) \end{aligned}$$

again, differentiate:

$$\partial_{\vec{M}} L = 2(\underline{L^T \cdot L}) \vec{M} - \underline{G^T \cdot \vec{\lambda}} = 0$$

yielding

$$\Rightarrow \vec{M}(\vec{\lambda}) = \frac{1}{2} (\underline{L^T \cdot L})^{-1} \underline{G^T \cdot \vec{\lambda}}. \text{ To determine } \vec{\lambda}, \text{ substitute}$$

back in:

$$\vec{d} = \underline{G} \cdot \vec{M} = \underline{G} \cdot \left(\frac{1}{2} (\underline{L^T \cdot L})^{-1} \underline{G^T \cdot \vec{\lambda}} \right)$$

$$\Rightarrow \vec{\lambda} = 2(\underline{G \cdot (L^T \cdot L)^{-1} \cdot G^T})^{-1} \cdot \vec{d}$$

Substituting the expression for λ into our expression for $\vec{M}$ yields:

$$\vec{M}(\vec{x}) = \frac{1}{2} (\underline{L}^T \cdot \underline{L})^{-1} \cdot \underline{G}^T \cdot \vec{J} = \frac{1}{2} (\underline{L}^T \cdot \underline{L})^{-1} \cdot \underline{G}^T \cdot (2 (\underline{G} (\underline{L}^T \cdot \underline{L})^{-1} \cdot \underline{G}^T)^{-1} \cdot \vec{d})$$

$$\Rightarrow \vec{M} = \underline{(\underline{L}^T \cdot \underline{L})^{-1} \cdot \underline{G}^T \cdot (\underline{G} \cdot (\underline{L}^T \cdot \underline{L})^{-1} \cdot \underline{G}^T)^{-1} \cdot \vec{d}}$$

This solution minimizes the rate of change, or "roughness" of the model, as prescribed by the first-order differential operator $\underline{L}$.

1) C. Now, we include a Lagrange Multiplier (λ) in our objective function (different from the vector factor λ in (2,3)!) that "weights" the model norm and the residuals against each other in the magnitude of the objective function. We also employ a target misfit equal to $\epsilon = 10 \text{ MV/M}$, since the stacking noise is 2-3 orders of magnitude weaker everywhere.

To determine the model which most accurately replicates the data (to within the target misfit) as well as contains a physically-motivated (i.e., linear) behavior between observation, we perform the inversion from (1b), equation (3), using many different Lagrange multipliers spanning from $\lambda \in [10^{-5}, 10^5]$.

The Figures that follow depict the various properties of our Fitting procedure for the target RMS misfit (see above) and with various values of λ . As indicated by the L-curve, the underdetermined nature of our procedure ~~does~~ combined with the $\|L_1 \cdot \tilde{m}\|$ Model norm does not yield the "nice" behavior of a global minimum distance to the origin at the target misfit. In other words, it is not an "L" shape. This is because the misfit remains high even as the $\|L_1 \cdot \tilde{m}\|$ norm is minimized, because a flat line cannot match the observations. In turn, with a low norm, the same "overfitting" that occurred in (1b) for the orange curve (Objective function (2)) will be produced by the model, yielding a very high norm. The "kink" or "elbow" in the curve occurs where the target misfit is achieved, as intended by the teal star. The other panel demonstrates that the RMS misfit follows the expected exponential (linear in log scale), then "flat" behavior with increasing λ value.

Next, we depict the minimum deviation from target misfit Model in teal. As expected, the solution is similar to our model obtained in part (b), but it also includes slightly more relaxation or dampening to within the target misfit range (error bars). Finally, we demonstrate all solutions obtained with our sweep over λ -space: Similar to the behavior indicated by the L-curve figure, the trade-off introduced by varying λ in our objective function produces "sharp" models that slightly over-fit or fit exactly (with low RMS misfit, which we know to be inaccurate) versus "flat" models that have minimal norms.

Tabulated values are included below:
(λ , RMS, $\|L_1 \cdot \tilde{m}\|$, $\|\tilde{m}\|$)

norm: L_1 , i.e.,
 $\|L_1 \cdot \tilde{m}\|$

Problem 2: Generalizing the inversion to a map

To perform the 2-D inversion, we generalized our 1-D approach from problem 1(c):

→ We consider a 1st-derivative model norm minimization or 1st-order Tikhonov regularizer, to obtain a physically appropriate "smoothness" of the $\vec{E}$ -field data

→ We employ the `scipy.kron` (Kronecker delta) function to generate the $\underline{L}_i$ arrays for each dimension

→ We use a locally-linear basis to construct $\underline{G}$, involving obtaining bi-linear weights according to

$$V_1 = \frac{1}{x_F y_F} (V(x, y)), \quad V_2 = \frac{1}{(1-x_F) y_F} \cdot V(x, y), \quad V_3 = \frac{1}{(1-x_F) x_F} \cdot V(x, y), \\ V_4 = \frac{1}{(1-x_F)(1-y_F)} \cdot V(x, y), \text{ for the schematic depicted in Fig 6}$$

of Constable et al., 2018.

→ We store all matrices as `scipy.sparse.csr_matrix` (i.e. sparse matrices) since the 100x100 discretization would otherwise cause large slowdown (memory utilization). We use a sparse matrix solving algorithm, `scipy.sparse.linalg.lsqr` (linear algebra module).

→ Again, the noise floor of 10 is much larger than the stacking errors, and dominates the uncertainty of the inversion. However, now the RMS misfit needs to be slightly smaller than the magnitude of the noise floor, since the problem is extremely sparse (few observations (626) compared to 10,000 points).

⇒ We use the L-curve analysis, following our logic from P1(c) (VDLS), to determine an appropriate choice of the noise floor. We select a target RMS misfit of $\frac{1}{2} \times \text{noise floor}$, or 5 MV/m. This occurs where the RMS misfit begins to grow but is not substantially high (compared to the noise floor), but also where the model norm $\|\underline{L}^T \cdot \underline{L} \cdot \vec{M}\|$ begins to drop linearly, it is far lower than its maximum value in λ -space.

(See attached figure)

→ Our solution produces most features of Constable et al., (2018), accounting for their sign change (see pg. 54 discussion). The features are sharper with a lower target misfit, but this is even farther below the noise floor than their work.